Multiple regression CHEAT SHEET

THE MODEL AND LEAST SQUARES

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \varepsilon$ (p x's)

$\hat{y} = b_0 + b_1 x_1 + \ldots + b_p x_p$ (b estimates β)

least squares: choose the b's to minimise $\Sigma(y - \hat{y})^2$

flats: $\hat{y} = 509.73 + 16.95$ size $- 40.49$ distance

70 m², 5 km → $\hat{y} \approx$ \$1,493 a month

LINEST

=LINEST(y, x_1:x_p, TRUE, TRUE): a 5-row block

row 1: $b_p \ldots b_1, b_0$ (REVERSED order)

row 2: their standard errors

row 3: R^2, s · row 4: F, $n - p - 1$ · row 5: SSR, SSE

INDEX(LINEST(…), row, col) pulls one value

=TREND(y, x's, new x's) predicts

HOLDING THE OTHERS FIXED

each b_i: change in $\hat{y}$ for one unit of x_i, the other x's held fixed

distance alone: +\$56.98 per km ($r^2 = 0.33$)

with size: −\$40.49 per km: the sign flips

why: r(size, distance) ≈ 0.80, bigger flats are farther

association in past data, not proof of cause

R SQUARED AND ADJUSTED R SQUARED

$SST = SSR + SSE$; $R^2 = SSR/SST$

flats: $R^2 = 3{,}642{,}774/4{,}043{,}299 = 0.90$

$$R^2_a = 1 - (1 - R^2)\frac{n-1}{n-p-1} = 0.896$$

adding any x can't lower R^2; R^2_a can fall

assumptions on ε: mean 0, variance σ^2 everywhere, independent, normal

THE F TEST AND T TESTS

F test $H_0: \beta_1 = \ldots = \beta_p = 0$; $F = \frac{MSR}{MSE} = \frac{SSR/p}{SSE/(n-p-1)}$

flats: $\frac{1{,}821{,}387}{10{,}825} = 168.3$, $p < 10^{-18}$

=F.DIST.RT(F, p, $n - p - 1$)

t test $H_0: \beta_i = 0$; $t = b_i/s_{b_i}$, df $n - p - 1$

distance: $-40.49/8.42 = -4.81$, $p \approx 0.00003$

=T.DIST.2T(|t|, $n - p - 1$); F first, then the t's

MULTICOLLINEARITY

x's strongly related: their effects can't be split

$VIF_j = 1/(1 - R_j^2)$ (x_j on the other x's); > 10 serious

rooms with size: t = −0.11 (p 0.92); alone with distance t ≈ 6.8

VIF(size) ≈ 8.0; s_b(size) 1.16 → 2.03

predictions inside the data: still fine

ESTIMATION AND PREDICTION

mean of y at x_0: $\hat{y} \pm t\, s_{\hat{y}}$

one new y at x_0: $\hat{y} \pm t\sqrt{(s^2 + s_{\hat{y}}^2)}$ (wider)

70 m², 5 km: CI \$1,456 … \$1,531

PI \$1,279 … \$1,708 (nearly 6 × as wide)

DUMMY VARIABLES

a category with k levels → $k - 1$ dummies (1/0)

the level without a dummy is the baseline

k dummies + intercept: perfect multicollinearity

parking: +\$75.11 (t 2.25, p 0.031), size and km fixed

a dummy shifts the plane; parallel planes

RESIDUAL ANALYSIS

plot e against $\hat{y}$: an even band, no curve, no funnel

standardized $e/(s\sqrt{(1 - h_i)})$: beyond ±2 ≈ 5 %

leverage $h_i > 3(p + 1)/n$: flat 24, 0.303 > 0.30

Cook's D > 1 is influential; max here 0.13

(software may give studentized deleted residuals)

Multiple regression CHEAT SHEET

LOGISTIC REGRESSION

$y = 1$ or 0: $p = \frac{\exp(b_0 + b_1 x)}{1 + \exp(b_0 + b_1 x)}$ $(0 < p < 1)$

odds $= p/(1-p)$; logit $= \ln(\text{odds}) = b_0 + b_1 x$

maximum likelihood (Python: statsmodels Logit)

loans: logit $= -4.84 + 0.0952$ DTI

odds ratio $e^{b_1} = 1.10$ a point; 2.59 per 10 points

DTI 40 → $p \approx 0.26$; DTI 20 → $p \approx 0.05$

guarantor: $e^{-2.60} = 0.074$ (odds −93 %)

$G = 2(LL - LL_0) = 14.9$, $\chi^2(1)$: $p \approx 0.0001$

an odds ratio multiplies odds, not probabilities

CURVES, INTERACTIONS AND LOGS

general linear model: linear in the b's

curve: $\hat{y} = b_0 + b_1 x + b_2 x^2$; turning point $-b_1/(2b_2)$

fuel: lowest at ≈ 57 km/h, ≈ 12.8 L/100 km (R^2 0.005 → 0.88)

interaction: $\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_1 x_2$

heating: $9.26 per degree day; insulated $5.19

log y: $\ln \hat{y} = b_0 + b_1 x$; $e^{b_1} - 1 =$ % per unit

forklifts: $e^{0.1108} - 1 = +11.7$ % a year

$y = \beta_0 \beta_1^x \rightarrow \ln y = \ln \beta_0 + x \ln \beta_1$

ADDING OR DELETING VARIABLES

$F = \frac{(\text{SSE(reduced)} - \text{SSE(full)})/q}{\text{MSE(full)}}$

flats + age, floor, balcony: $\frac{54{,}217/3}{297{,}085/33} = 2.01$

=F.DIST.RT(2.01, 3, 33) = 0.13: leave them out

a category's dummies go in or out together

VARIABLE SELECTION

forward: add the smallest p while $p < 0.05$

backward: drop the largest p while $p > 0.05$

stepwise: forward, with a check to leave

best subsets: all $2^7 - 1 = 127$ models, by R^2_a

flats: size, distance, parking (R^2 0.913)

danger: 20 noise x's at 0.05 → $1 - 0.95^{20} \approx 64$ %

test the final model on held-back data

one-way ANOVA = regression on k − 1 dummies (F 14.44)

AUTOCORRELATION: DURBIN–WATSON

$d = \frac{\Sigma(e_t - e_{t-1})^2}{\Sigma e_t^2}$ = SUMXMY2(e_2:e_n, e_1:e_{n-1})/SUMSQ(e)

0 … 4; near 2: none; well below 2: positive

$d < d_L$ reject · $d > d_U$ don't · between: inconclusive

α 0.05: n 24, one x 1.273 / 1.446; n 30, one x 1.352 / 1.489

supplier: $d = 0.83 < 1.273$: autocorrelated;

on the monthly changes $d \approx 2.02$

CASE: CORROWBY FLEET LEASING (FICTIONAL)

240 vans: age, thousand km, fuel, condition, price

good vs fair: $26,155 vs $18,715: mostly age

linear price: residuals curve and fan out → ln(price)

age −9.8 %/yr · per 1,000 km −0.15 % · electric +12.4 %

good: +10.7 % (95 %: 7.8 % … 13.6 %)

$1,500 pays above $14,030 (mid), $19,190 (cautious)

→ refurbish fair vans predicted > ≈ $19,200; trial first

SPREADSHEET AND PYTHON

LINEST(y, x's, TRUE, TRUE) · INDEX · TREND

F.DIST.RT(F, df1, df2) · T.DIST.2T(|t|, df)

T.INV.2T(α, df) · CORREL · RSQ · SUMPRODUCT

LN · EXP · SUMSQ · SUMXMY2 · AVERAGEIFS

Python: numpy.linalg.lstsq; statsmodels OLS, Logit,

variance_inflation_factor, durbin_watson